KDU FACULTY OF COMPUTING · INDIVIDUAL RESEARCH PROJECT (CS4999)

# Computational Modeling of Cognitive Performance & Gaming Behavior

Progress Review II Technical Deliverables & Evaluation Summary


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**Current Status:** All Deliverables Deployed Successfully

## 1. Project Synopsis & Objectives

Gaming disorder is an emerging psychological threat in student populations. This project develops a **unified, multi-modal machine learning system** integrating clinical self-reports, cognitive biometrics, and physical fatigue metrics to accurately identify addiction risk and trace cognitive performance levels. To fulfill the requirements of Progress Review II, we have delivered a fully integrated interactive dashboard utilizing state-of-the-art ensemble models, deep CNN models, and automated reporting.

## 2. Technical Machine Learning & Evaluation Leaderboard

A rigorous classifier exploration was completed by training, tuning, and comparing 14 pipeline configurations. The models are evaluated below using Stratified 5-Fold Cross-Validation to ensure generalization:

| Rank | Algorithm / Configuration | Test Acc | CV Acc (5-Fold) | CV Std | Selection Status |
|---|---|---|---|---|---|
| 1 | **Stacking Ensemble** | 85.00% | 83.00% | ±4.40% | ■ **Selected Best** |
| 2 | Random Forest (Base) | 80.00% | 82.00% | ±5.52% | Superseded |
| 3 | Random Forest (Tuned) | 81.67% | 81.67% | ±3.16% | Candidate |
| 4 | Extra Trees | 76.67% | 81.33% | ±3.86% | Candidate |
| 5 | XGBoost (Tuned) | 78.33% | 81.00% | ±5.01% | Candidate |
| 6 | Voting Ensemble | 80.00% | 81.00% | ±4.03% | Candidate |
| 7 | LightGBM | 76.67% | 78.33% | ±2.58% | Candidate |
| 8 | Gradient Boosting | 83.33% | 78.00% | ±5.31% | Candidate |
| 9 | AdaBoost | 73.33% | 74.67% | ±5.31% | Baseline |
| 10 | SVM (RBF) | 76.67% | 68.67% | ±2.87% | Baseline |
| 11 | Decision Tree | 66.67% | 67.00% | ±3.71% | Baseline |
| 12 | MLP Neural Net | 48.33% | 60.67% | ±10.78% | Baseline |

## 3. Multi-Modal Regressions & Biometric Performance

- **Deep ResNet50 Eye Fatigue Classifier:** Trained on 84,898 eye images (MRL dataset). Reached **66.33% accuracy** with an **AUC of 0.89**. Demonstrates a high fatigue detection recall rate of 93.00% to safeguard screening parameters.
- **Attention score Regressor:** Leverages Random Forest regression models to predict student focus. Accomplished an improved $R^2$ score of **0.417** (RMSE: 0.099).

- **Memory score Regressor:** Predicts short-term recall capacities. Accomplished an $R^2$ score of **0.176** (RMSE: 0.135).
- **Late Fusion Integration:** Aggregates classifiers, biometrics, and K-Means predictions using a confidence-weighted late fusion rule, achieving a 3.00% performance boost over baseline architectures.

## 4. Implemented Novelty Features (App v5 Research Edition)

To maximize research contribution and multi-modal integration, the Streamlit app was upgraded to Version 5 (Research Edition) featuring 6 core integrated modules:

1. **Multimodal Late Fusion Engine:** Combines Tabular Stacking Ensemble, Cognitive ANN, and CNN Eye Fatigue prior using confidence-weighted log-linear opinion pooling.
2. **Live Webcam & Computer Vision Eye Scanner:** Captures live eye photos via OpenCV, extracting Aperture Openness, Scleral Redness, Brightness, and Contrast.
3. **Interactive 'What-If' Sensitivity Simulator:** Real-time habit lab showing how lifestyle modifications affect Addiction Risk %, Health Score, and Recovery Days.
4. **6-Axis Gamer Persona Radar Chart:** Spider-web visualization comparing individual behavioral & cognitive metrics against gamer population benchmarks.
5. **20-20-20 Eye Strain & Detox Planner:** Clinical ocular fatigue prevention routines, recovery timeline estimators, and daily detox schedules.
6. **Upgraded ReportLab PDF Exporter:** Enables users to download a signed, structured verification PDF summary report directly from the web dashboard.

## 5. Codebase Cleanup & System Architecture

A full directory refactoring was completed. Obsolete scripts and temporary stubs were safely cleaned. The project is consolidated under structured directories (data/, models/, notebooks/, and results/). The main application is deployed via app_v5.py using a virtualized .venv.

**Report Certified By: Final Year Research Project Team | Intake 40**